

Conflict-Aware Cross-View Disaster Triage Across Spatial Observation Regimes [experiments]

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ABSTRACT

Cross-view disaster assessment combines ground-level and overhead imagery to infer building damage. Aggregate F1 scores, however, hide the cases where fusion matters most: examples for which the single-view predictors disagree. This short paper introduces a Conflict-Aware Evaluation protocol (CAE) for paired-view disaster triage. CAE reports overall F1, conflict-subset F1, and the conflict gain of cross-view fusion over the stronger single-view baseline. We apply CAE to two paired disaster settings: an Eaton/Altadena wildfire dataset with property-centered ground imagery and the Hurricane Ian CVDIsaster benchmark with panoramic ground imagery. The results show that cross-view fusion is most informative on disagreement cases, but the dominant single-view arbitrator depends on the observation regime and the split definition. On the wildfire sensitivity split, cross-view fusion improves conflict F1 by 0.0388 over the best single-view baseline. On a leakage-controlled hurricane endpoint split, cross-view fusion no longer dominates street-only inference on conflict cases, which provides an important boundary condition. We further report two pilot extensions: conflict density as an unsupervised tile-level wildfire damage proxy and a conflict-weighted loss that improves wildfire conflict F1. The evidence supports a conservative conclusion: paired-view disaster models should be evaluated as conflict resolvers, not only as average-case classifiers.

KEYWORDS

disaster assessment, cross-view learning, multimodal fusion, GeoAI, conflict-aware evaluation

1 INTRODUCTION

Post-disaster building assessment increasingly uses several visual viewpoints. Overhead imagery gives broad spatial coverage, while ground imagery can expose facade condition, debris, and local structural cues. Cross-view models seek to combine these signals, but prior evaluation often reduces their value to one aggregate test metric. That view is too coarse for operational disaster triage. Many buildings are easy in both views. The harder and more consequential cases are those where the two views provide competing evidence.

This paper studies cross-view fusion as a conflict-resolution problem. We focus on paired examples that contain a ground image, an overhead patch, and a binary damage label. Instead of asking only whether paired-view fusion improves average F1, we ask when fusion helps after the two single-view models disagree. That framing matters for geographic AI because observation conditions are not interchangeable. A close property inspection photograph and a panoramic street scene both count as ground imagery, but they

encode different relationships between observer, target building, and surrounding landscape.

The study sits between several lines of disaster-vision work. Satellite damage benchmarks such as xBD have made overhead-only building assessment a standard task [1], and later fusion models improve satellite-based damage assessment from multi-scale visual evidence [5]. Ground and street-view studies show that close-range imagery can capture damage cues that overhead imagery misses [6–8]. CVDIsaster is the closest paired-view benchmark for Hurricane Ian [3]. CAE adds a different evaluation axis by separating average performance from behavior on single-view disagreement cases.

We contribute a compact evaluation study for ACM SIGSPATIAL short-paper review. First, we define Conflict-Aware Evaluation (CAE), a three-metric protocol that reports full-test performance and conflict-subset behavior together. Second, we apply CAE to wildfire and hurricane paired-view triage, while treating the hurricane grouped-clean split as the conservative endpoint control. Third, we show how the same disagreement signal can support two extensions: wildfire tile-level conflict-density mapping and a conflict-weighted training loss. The paper deliberately avoids a causal claim that disaster type alone determines which view should dominate. The evidence is better read as a regime-sensitive experiment: cross-view fusion helps most when single-view evidence conflicts, and the source of that help depends on how the ground view observes the target.

2 DATA AND PROTOCOL

2.1 Paired-View Triage Settings

Let $\mathcal{D} = \{(x_i^{(s)}, x_i^{(r)}, y_i)\}_{i=1}^N$ be a paired disaster triage dataset, where $x_i^{(s)}$ is a street or ground image, $x_i^{(r)}$ is an overhead patch, and $y_i \in \{0, 1\}$ is the binary damage label. We compare three modes: *street-only*, *remote-only*, and *cross-view*. The cross-view model uses late fusion of ResNet18 features [2], including each embedding, their absolute difference, and their elementwise product. This simple fusion design keeps the evaluation focused on conflict behavior rather than architectural novelty.

The wildfire setting uses an Eaton/Altadena paired dataset assembled for this project. Its ground imagery is property-centered inspection imagery paired with overhead patches. This observation regime often places the target building near the visual center of the ground image. The hurricane setting uses the Hurricane Ian CVDIsaster benchmark [3], which pairs panoramic ground imagery with very high resolution overhead imagery. This observation regime often includes wider street context, adjacent buildings, and weaker

Table 1: Conflict-Aware Evaluation summary. Conflict F1 uses C_τ with $\tau = 0.1$. WF is wildfire; H is hurricane. Gain is reported only for cross-view rows.

Setting	M	All	C-F1	Gain	n_c
WF sens.	S	0.9392	0.6230	–	435
WF sens.	R	0.9238	0.5237	–	435
WF sens.	CV	0.9473	0.6618	+0.0388	435
H end.	S	0.9055	0.6842	–	32
H end.	R	0.8969	0.5806	–	32
H end.	CV	0.9192	0.7647	+0.0805	32
H mod.+sev.	S	0.8738	0.7324	–	128
H mod.+sev.	R	0.8873	0.7742	–	128
H mod.+sev.	CV	0.8915	0.7867	+0.0125	128
H clean	S	0.9016	0.8101	–	80
H clean	R	0.8385	0.6526	–	80
H clean	CV	0.9008	0.7949	-0.0153	80

target alignment. The comparison therefore mixes disaster type, geography, source, and collection protocol; those factors limit causal interpretation and motivate the conservative language used below.

2.2 Conflict-Aware Evaluation

For each example, let $p_i^{(s)}$ and $p_i^{(r)}$ denote the positive-class probabilities predicted by the street-only and remote-only models. CAE defines the conflict subset

$$C_\tau = \{i : |p_i^{(s)} - p_i^{(r)}| > \tau\}, \quad (1)$$

with $\tau = 0.1$ in the main analysis. For each model m , CAE reports standard overall F1 and conflict F1 on C_τ . It then computes the cross-view conflict gain

$$\Delta_{\text{view}} = \text{F1}(\text{crossview}; C_\tau) - \max\{\text{F1}(\text{street}; C_\tau), \text{F1}(\text{remote}; C_\tau)\}. \quad (2)$$

This protocol exposes whether paired-view fusion resolves disagreement cases rather than only improving easy average cases.

We also decompose conflicts by correction direction. $S \rightarrow O$ denotes cases where the street-only model is correct and the remote-only model is wrong; $O \rightarrow S$ denotes the reverse. These fractions identify the stronger single-view arbitrator inside the conflict subset. Because a prior hurricane endpoint split had object-level leakage, we report the grouped-clean endpoint split as a required control rather than as an optional robustness run.

3 RESULTS

3.1 CAE Summary

Table 1 reports the core CAE results. The wildfire sensitivity split has the clearest positive conflict gain: cross-view fusion reaches 0.6618 conflict F1, which is 0.0388 above the better single-view baseline. The original hurricane endpoint split also shows a large gain, but its small conflict subset ($n = 32$) and leakage history make it diagnostic rather than primary evidence. The broader hurricane moderate-plus-severe split keeps a smaller positive gain. The grouped-clean endpoint split provides the most important boundary condition: cross-view fusion remains close to street-only overall, but it no longer dominates street-only on the conflict subset.

The directional decomposition sharpens this interpretation. On wildfire, $S \rightarrow O$ conflicts are more common than $O \rightarrow S$ conflicts (0.2713 versus 0.1885), so the ground image is the stronger single-view arbitrator. On the hurricane endpoint split, the two directions are

Table 2: Pilot method and spatial extensions.

Extension	Primary statistic	Scope
Conflict density	$r = 0.5123, p = 0.0063$	wildfire tiles
ConflictFocalLoss	conflict F1 = 0.7208	wildfire split
Adaptive cascade	F1 = 0.9498, 9.5% routed	wildfire split

near tied (0.3750 versus 0.3438). On the broader hurricane split, the ordering reverses (0.1094 versus 0.1328), making the overhead view the stronger single-view arbitrator. The stable claim is therefore not that one disaster always favors one modality. It is that conflict behavior changes with the observation regime and split definition.

3.2 Spatial and Training Extensions

CAE also produces useful secondary signals. For the wildfire dataset, we aggregate $|p_i^{(s)} - p_i^{(r)}|$ by overhead tile and compare the resulting conflict density with observed tile damage rate. Across 27 tiles, Spearman $r = 0.5123$ and $p = 0.0063$. This suggests that cross-view disagreement can act as an unsupervised spatial damage proxy in the property-centered wildfire setting. We do not yet claim the same result for hurricane because the benchmark does not provide equivalent tile metadata in this repository.

We also test ConflictFocalLoss, a conflict-weighted binary loss related in spirit to focal reweighting [4]. For each cross-view example, the loss multiplies binary cross entropy by $(1 + c_i)$, where c_i is a feature-space disagreement score based on the cosine distance between the two view embeddings. On the wildfire sensitivity split, moderate conflict weighting improves overall F1 from 0.9473 to 0.9520 and conflict F1 from 0.6618 to 0.7208. This result is encouraging, but it is still a single-setting pilot rather than a fully validated method contribution.

4 DISCUSSION AND LIMITATIONS

The experiment supports CAE as a useful evaluation protocol for paired-view disaster triage. Aggregate F1 alone would make the models look similar in some settings. CAE reveals how they behave on disagreement cases and shows that cross-view value can shrink, grow, or even become negative relative to the stronger single-view model after leakage control.

Several limits are central. First, the wildfire and hurricane settings differ by more than observation regime. They also differ by hazard, geography, image source, label construction, and metadata availability. The results therefore support a regime-sensitive interpretation, not a causal proof. Second, the hurricane endpoint conflict subset is small, so its large positive gain should be treated as tentative. Third, the private wildfire data package must include complete counts, class balance, annotation protocol, inter-annotator agreement if available, and release or access instructions before final submission. Fourth, ConflictFocalLoss and the tile-level spatial proxy are validated only on the wildfire setting. They motivate follow-up experiments but should not be presented as fully general.

These limits also clarify the paper's contribution. CAE does not claim that cross-view fusion always wins. It gives reviewers and

practitioners a clearer way to see when paired views resolve evidence conflict and when a single view remains sufficient.

5 CONCLUSION

This short paper reframes paired-view disaster triage as conflict-aware evaluation. Across wildfire and hurricane paired-view settings, cross-view fusion is most informative on disagreement cases, but the stronger single-view arbitrator changes with observation regime, split definition, and leakage control. The resulting lesson is practical: disaster benchmarks should report conflict-subset behavior alongside aggregate scores. Doing so makes multimodal fusion more falsifiable, more spatially interpretable, and more useful for deployment decisions.

ACKNOWLEDGMENTS

The authors used AI-based writing assistance to support grammar checking, draft compression, and editorial revision. These tools were not used as sources of scientific evidence or to autonomously generate the study design, datasets, experimental results, figures, tables, interpretations, or conclusions. The authors reviewed and approved all manuscript content and take full responsibility for the work.

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